

Lattice Quantization for Smooth Compression: Euclidean and KL Quantizers

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Abstract

Small edits to raw data can cause large changes in the compressed version, which can be costly in applications such as data synchronization and incremental backup. We define smoothness as the ability to keep changes in the compressed version small under small edits to the raw data. In this paper, we first quantize the empirical distribution and then construct a Huffman code from the quantized distribution, which can improve smoothness at the cost of some compression efficiency.

We study two quantizers based on a shifted and scaled lattice in the probability simplex: the Euclidean Lattice Quantizer (ELQ) and the KL Quantizer (KLQ). For ELQ, we derive bounds on smoothness and KL distortion. For KLQ, we show that its quantization cells can be determined by comparisons with adjacent lattice codewords and use the distances to the cell boundaries to derive a smoothness bound. For a symmetric codebook of positive types, we obtain the exact worst-case KL distortion over the full probability simplex. We also give a heuristic choice of the minimum codeword probability. Simulations using canonical Huffman coding compare ELQ and KLQ with an existing compander-based quantizer and show the trade-off between KL distortion and codebook stability under random deletions.

1 Introduction

Lossless compression reduces storage and communication costs while allowing the original data to be recovered exactly. Methods such as Huffman coding and arithmetic coding [1, 2] use source probabilities to reduce the number of bits needed to encode the data. In incremental backups and data synchronization, a small change to a file may still require a large update to its compressed version [3, 4]. This can increase the amount of data that needs to be transferred or rewritten. For Huffman coding, a small edit can change the empirical distribution and cause many codewords to change when the code is rebuilt [5]. Even symbols that were not edited may then be assigned different codewords, increasing the number of bits that change in the compressed version.

Smooth compression asks that small edits to the input cause only small changes in the compressed version. Montanari and Mossel [6] defined smoothness using a Lipschitz condition on Hamming distance and constructed smooth codes based on sparse graphs. Vatedka and Tchamkerten [7] studied a related problem of local decoding and update, where a small part of the source can be read or updated by accessing

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only a small number of bits in the compressed data. Our previous work [5] focused on Huffman coding and studied how its codebook changes under edits. We showed that canonical Huffman coding is more stable than standard Huffman coding and proposed a pre-allocation method that further improves stability while keeping the Huffman code optimal. Varshney et al. [8] studied another setting that trades some compression efficiency for a smaller edit distance between compressed versions.

One way to make this trade-off is to quantize the source distribution before coding. Our recent work [9] studied this approach. Nearby empirical distributions may map to the same quantized distribution and use the same codebook. We proposed a weighted geometric quantizer and a covering-code method over the type space, and studied their KL distortion and code stability under edits. Probability distribution quantization has also been studied independently of smooth compression. Reznik [10] studied Euclidean nearest-neighbor quantization on type lattices and biased types whose entries remain positive. Geiger and Böcherer [11] studied M -type approximation and gave an exact greedy algorithm for minimizing KL divergence over all M -types. Adler et al. [12] studied a scalar quantization method for probability distributions and analyzed its KL distortion.

In this paper, we study whether lattice quantization can provide a good trade-off between smoothness and KL distortion. We use a shifted and scaled A_{m-1} lattice inside the probability simplex, whose codebook is a special case of the biased types studied by Reznik [10]. On this codebook, the Euclidean Lattice Quantizer (ELQ) maps a distribution to the nearest codeword in Euclidean distance, while the KL Quantizer (KLQ) chooses the codeword that minimizes $D(\mathbf{p} \parallel \mathbf{q})$. We measure smoothness by the probability that the quantized distribution remains unchanged after a small random perturbation. For ELQ, we derive bounds on smoothness and KL distortion. For KLQ, we show that each cell can be determined by comparisons with lattice-adjacent codewords and use the distances to the cell boundaries to derive a smoothness bound. For the symmetric codebook consisting of all positive types of length K , we also obtain the exact worst-case KL distortion over the full probability simplex, $\ln(K/(K - m + 1))$. We give a heuristic choice of the minimum codeword

Table 1: Comparison of KL-distortion results with prior work.

Setting	KL distortion
ELQ unclipped cell, $\min_i q_i \geq \eta > 0$	Prior: $D = O(n^{-2})$ [10] Ours: $D \leq \frac{m\delta^2}{4\eta}$
KLQ $p \in \Delta_{m-1}^\circ$, $p \rightarrow e_j$	Prior upper bound: $\rightarrow \ln \frac{K}{K - m + 1} + \frac{\ln 2}{2(K - m + 1)}$ [11] Ours: $D \rightarrow \ln \frac{K}{K - m + 1}$

probability α . Table 1 compares our KL-distortion results with prior work under the same settings.

For ELQ, with $n = km$ and $\beta = \alpha/\delta$ in Reznik's notation, fixing m , η , and β gives $\delta = \Theta(n^{-1})$, so our bound has the same $O(n^{-2})$ rate and is nonasymptotic. For KLQ, our near-boundary limit gives the exact global worst-case distortion. Neither of these prior works studies smoothness, while we give smoothness bounds for both ELQ and KLQ.

In simulations, we use canonical Huffman coding after quantization by ELQ, KLQ, and the method of Adler et al. [12], and compare them with standard Huffman and the unquantized canonical baseline. We measure both KL distortion and the probability that the Huffman codebook remains unchanged after random deletions.

2 Preliminaries

Let $m \geq 2$ be an integer. The $(m-1)$ -simplex is $\Delta_{m-1} = \{\mathbf{p} \in \mathbb{R}_{\geq 0}^m : \sum_i p_i = 1\}$, with interior Δ_{m-1}° and tangent subspace $H_0 = \{\mathbf{x} : \sum_i x_i = 0\}$. The A_{m-1} root lattice $\{\mathbf{z} \in \mathbb{Z}^m : \sum_i z_i = 0\}$ has minimal vectors $\mathbf{e}_i - \mathbf{e}_j$ of length $\sqrt{2}$. Figure 1a shows the $m = 3$ case.

A *type* of length $K \geq m$ is a distribution on $\{1, \dots, m\}$ with entries in $\frac{1}{K}\mathbb{Z}$. Let $\mathring{\mathbb{S}}_K^m = \{\mathbf{p} \in \Delta_{m-1}^\circ : Kp_i \in \mathbb{Z}_{\geq 1} \ \forall i\}$ denote the *interior types*, and let $\hat{\mathbb{S}}_K^m = \{\mathbf{p} \in \Delta_{m-1} : Kp_i \in \mathbb{Z}_{\geq 0} \ \forall i, \min_i p_i = 0\}$ denote the *boundary types*.

For $\mathbf{p}, \mathbf{q} \in \Delta_{m-1}$, the total variation distance is $\text{TV}(\mathbf{p}, \mathbf{q}) = \frac{1}{2}\|\mathbf{p} - \mathbf{q}\|_1$. When $q_i > 0$ for every i with $p_i > 0$, the KL divergence is $D(\mathbf{p} \parallel \mathbf{q}) = \sum_{i:p_i > 0} p_i \ln \frac{p_i}{q_i}$. We will use the bound

$$D(\mathbf{p} \parallel \mathbf{q}) \leq \sum_{i:p_i > 0} p_i \ln \left(\max_{j:p_j > 0} \frac{p_j}{q_j} \right) = \ln \max_{i:p_i > 0} \frac{p_i}{q_i}. \quad (1)$$

A *quantizer* on Δ_{m-1} has a finite *codebook* $\mathcal{C} \subset \Delta_{m-1}$ and cells $\{V(\mathbf{q})\}_{\mathbf{q} \in \mathcal{C}}$ that partition Δ_{m-1} . The map $\phi : \Delta_{m-1} \rightarrow \mathcal{C}$ sends $\mathbf{p}$ to the codeword $\mathbf{q}$ whose cell contains $\mathbf{p}$.

For a perturbation size ε , the smoothness of a quantizer ϕ is

$$S(\varepsilon) = \Pr(\phi(\mathbf{p}) = \phi(\mathbf{p}')), \quad (2)$$

where $\mathbf{p} \sim \text{Unif}(\Delta_{m-1})$ and, given $\mathbf{p}$, $\mathbf{p}'$ is drawn uniformly from the TV-ball of radius ε around $\mathbf{p}$.

Define $\hat{S}_2(\varepsilon)$ as the probability that a uniformly random $\mathbf{p} \in \Delta_{m-1}$ is at Euclidean distance at least 2ε from the boundary of its cell. All Euclidean distances to cell boundaries are taken within the affine hull of Δ_{m-1} . Since $\|\mathbf{p} - \mathbf{p}'\|_2 \leq 2\text{TV}(\mathbf{p}, \mathbf{p}') \leq 2\varepsilon$, such a point $\mathbf{p}$ remains in the same cell after the perturbation. Hence

$$S(\varepsilon) \geq \hat{S}_2(\varepsilon). \quad (3)$$

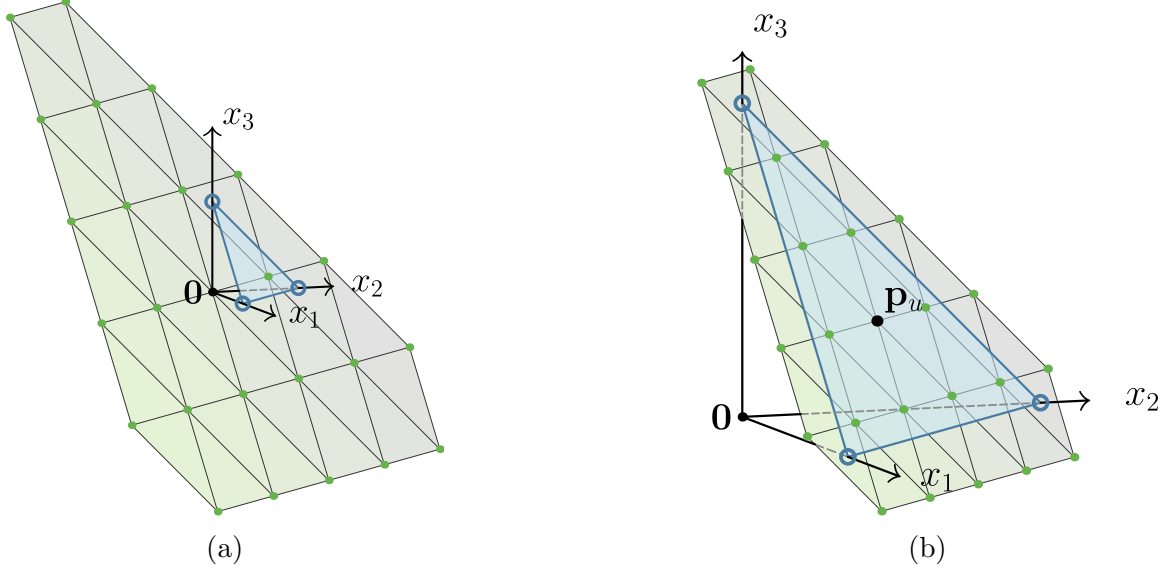


Figure 1: (a) The 2-simplex Δ_2 (corners shown as open rings, intersecting each axis at 1) and the A_2 root lattice (filled circles) in the parallel zero-sum plane through the origin. (b) The codebook $\mathcal{C}_\delta$ (Section 3) and surrounding lattice $\mathcal{L}_\delta = \mathbf{p}_u + \delta A_{m-1}$, centered at the labeled simplex centroid $\mathbf{p}_u = (1/3, 1/3, 1/3)$, for $m = 3$, $k = 1$, $\delta = 1/4$.

3 Codebook

Both quantizers we studied in this paper use the same lattice-based codebook, namely a shifted and scaled version $\mathcal{L}_\delta$ of the A_{m-1} lattice, restricted to the interior of the probability simplex Δ_{m-1} . Fix $\delta \in (0, 1/m)$ and let $\mathbf{p}_u = \mathbf{1}/m$ be the uniform distribution. Define

$$\mathcal{L}_\delta \triangleq \mathbf{p}_u + \delta A_{m-1}, \quad \mathcal{C}_\delta \triangleq \mathcal{L}_\delta \cap \Delta_{m-1}^\circ.$$

We exclude codewords on the boundary so that the KL divergence to any codeword is finite. An example for $m = 3$ is shown in Figure 1b. For later use, we write this codebook in terms of α , δ , and k .

Lemma 1. *For $\delta \in (0, 1/m)$, we have*

$$\mathcal{C}_\delta = \left\{ \alpha \mathbf{1} + \delta \mathbf{w} : \mathbf{w} \in \mathbb{Z}_{\geq 0}^m, \sum_i w_i = km \right\}, \quad (4)$$

where

$$k = \left\lceil \frac{1}{m\delta} \right\rceil - 1, \quad \alpha = \frac{1}{m} - k\delta \in (0, \delta]. \quad (5)$$

Furthermore, $|\mathcal{C}_\delta| = \binom{km+m-1}{m-1}$.

Proof. Any $\mathbf{q} \in \mathcal{L}_\delta$ has the form $\mathbf{q} = \mathbf{p}_u + \delta \mathbf{z}$ with $\mathbf{z} \in A_{m-1}$. Since $\mathbf{p}_u = (\alpha + k\delta)\mathbf{1}$, setting $\mathbf{w} = \mathbf{z} + k\mathbf{1}$ gives $\mathbf{q} = \alpha \mathbf{1} + \delta \mathbf{w}$ and a bijection between A_{m-1} and $\{\mathbf{w} \in \mathbb{Z}^m : \sum_i w_i = km\}$. Also, $q_i = \alpha + \delta w_i > 0$ implies $w_i > -\alpha/\delta$. The definition of k gives $\alpha \in (0, \delta]$, so $-\alpha/\delta \in [-1, 0)$. Thus, over the integers, $q_i > 0$ is equivalent to $w_i \geq 0$. This gives (4), and the cardinality follows by stars-and-bars. $\square$

This codebook is a special case of the biased-type construction of Reznik [10]. In his notation, $q_i = (w_i + \beta)/(n + m\beta)$, which matches our parametrization with $n = km$ and $\beta = \alpha/\delta$.

Here, k is the number of codewords strictly between $\mathbf{p}_u$ and any simplex vertex $\mathbf{e}_i$. Figure 2 shows an example with $m = 3$ and $k = 2$. The parameter α is the smallest value any coordinate of a codeword can take. The *corner* vectors, which are closest to the simplex vertices $\mathbf{e}_i$, are

$$\mathbf{q}_i = (\alpha, \dots, \alpha, \underbrace{\alpha + km\delta}_{\text{position } i}, \alpha, \dots, \alpha), \quad i = 1, \dots, m,$$

where $\alpha + km\delta = 1 - (m - 1)\alpha$ since the coordinates sum to 1.

For this codebook, set $K \triangleq m(k + 1)$. Then Lemma 1 gives

$$|\mathcal{C}_\delta| = \binom{K - 1}{m - 1} = |\mathring{\mathbb{S}}_K^m|. \quad (6)$$

For fixed k , the valid range is $\alpha \in (0, 1/K]$. The set equality $\mathcal{C}_\delta = \mathring{\mathbb{S}}_K^m$ requires the symmetric choice $\alpha = \delta = 1/K$, where the lattice spacing equals the boundary offset, as shown in Figure 2b. In Reznik's biased-type notation, this choice corresponds to $n = K - m$ and $\beta = 1$. In this case, lattice points on the simplex boundary are excluded from the codebook.

4 Euclidean Lattice Quantizer (ELQ)

In the previous section, we introduced the lattice-based codebook $\mathcal{C}_\delta$. In this section, we consider the Euclidean quantizer based on this codebook, which quantizes distributions in Δ_{m-1} to the nearest codeword in $\mathcal{C}_\delta$ using the Euclidean distance. Reznik [10] also studied Euclidean nearest-neighbor quantization on type lattices. Here we use the biased interior codebook $\mathcal{C}_\delta$ and focus on smoothness under distribution perturbations.

4.1 Voronoi Cells of the Quantizer

Definition 2. *The Euclidean Lattice Quantizer is*

$$\phi_{\text{ELQ}}(\mathbf{p}) \in \arg \min_{\mathbf{q} \in \mathcal{C}_\delta} \|\mathbf{p} - \mathbf{q}\|_2,$$

with ties broken arbitrarily. Its cells are $V_{\text{ELQ}}(\mathbf{q}) = \{\mathbf{p} \in \Delta_{m-1} : \|\mathbf{p} - \mathbf{q}\|_2 \leq \|\mathbf{p} - \mathbf{q}'\|_2 \ \forall \mathbf{q}' \in \mathcal{C}_\delta\}$.

Each pairwise Euclidean comparison defines a half-space, so every $V_{\text{ELQ}}(\mathbf{q})$ is a convex polytope. Figure 2 shows examples of the Voronoi cells for different choices of α . The choice $\alpha = \delta$ is easier to analyze. We discuss a heuristic choice of α in Section 4.4.

For the unrestricted lattice $\mathcal{L}_\delta \supset \mathcal{C}_\delta$, the Voronoi cell at $\mathbf{q}$ has vertices $\mathbf{q} + \delta\sigma(\mathbf{u}^{(j)})$, where σ ranges over coordinate permutations, $j \in \{1, \dots, m-1\}$, and

$$\mathbf{u}^{(j)} = \underbrace{(j/m, \dots, j/m)}_{m-j}, \underbrace{(j-m)/m, \dots, (j-m)/m)}_j, \quad (7)$$

with $\|\mathbf{u}^{(j)}\|_2^2 = j(m-j)/m \leq m/4$ [13, §21.3.B]. So every $\mathbf{p}$ in the unrestricted lattice Voronoi cell of $\mathbf{q}$ satisfies $\|\mathbf{p} - \mathbf{q}\|_2 \leq \frac{1}{2}\delta\sqrt{m}$.

4.2 Inradius and smoothness

We first lower-bound the ELQ cell inradius and then use this to lower-bound $\widehat{S}_2(\varepsilon)$ and hence the smoothness $S(\varepsilon)$.

Proposition 3. *For every $\mathbf{q} \in \mathcal{C}_\delta$, the inradius $\rho_{\text{ELQ}}(\mathbf{q}) := \text{dist}(\mathbf{q}, \partial V_{\text{ELQ}}(\mathbf{q}))$ satisfies*

$$\rho_{\text{ELQ}}(\mathbf{q}) \geq \min\left(\frac{\delta}{\sqrt{2}}, \alpha\sqrt{\frac{m}{m-1}}\right),$$

with equality whenever $\mathbf{q}$ has a lattice-adjacent neighbour in $\mathcal{C}_\delta$ and the minimum is attained by the first term (i.e., $\alpha \geq \delta\sqrt{\frac{m-1}{2m}}$).

Proof. By Definition 2, the cell $V_{\text{ELQ}}(\mathbf{q})$ is the intersection of the bisector half-spaces for all $\mathbf{q}' \in \mathcal{C}_\delta \setminus \{\mathbf{q}\}$ and the simplex Δ_{m-1} . The inradius $\rho_{\text{ELQ}}(\mathbf{q})$ is the minimum ℓ_2 -distance from $\mathbf{q}$ to any of these bounding hyperplanes.

For the bisectors, the distance to the perpendicular bisector of $\mathbf{q}'$ is $\|\mathbf{q} - \mathbf{q}'\|_2/2$. A lattice-adjacent neighbour $\mathbf{q}' = \mathbf{q} + \delta(\mathbf{e}_a - \mathbf{e}_b)$ gives $\|\mathbf{q} - \mathbf{q}'\|_2 = \delta\sqrt{2}$, so its bisector is at distance $\delta/\sqrt{2}$. Any non-adjacent neighbour $\mathbf{q}''$ has $\|\mathbf{q} - \mathbf{q}''\|_2 \geq 2\delta$, so its bisector is at distance at least δ . Thus, the minimum bisector distance is $\delta/\sqrt{2}$ whenever a lattice-adjacent neighbour exists.

For the simplex facets, within the affine hyperplane $H = \{\mathbf{p} : \sum_j p_j = 1\}$, the normal vector to the facet $\{p_i = 0\}$ is $\mathbf{e}_i - \frac{1}{m}\mathbf{1}$. Moving from $\mathbf{q}$ by $-\frac{q_i}{1-(1/m)}(\mathbf{e}_i - \frac{1}{m}\mathbf{1})$ reaches this facet. Its Euclidean length is $q_i\sqrt{\frac{m}{m-1}} \geq \alpha\sqrt{\frac{m}{m-1}}$. Taking the minimum over the bisectors and simplex facets gives $\rho_{\text{ELQ}}(\mathbf{q}) \geq \min(\frac{\delta}{\sqrt{2}}, \alpha\sqrt{\frac{m}{m-1}})$. $\square$

Theorem 4 (ELQ smoothness). *Let $\rho = \min(\frac{\delta}{\sqrt{2}}, \alpha\sqrt{\frac{m}{m-1}})$. For all $\varepsilon \in [0, \rho/2]$,*

$$S(\varepsilon) \geq \widehat{S}_2(\varepsilon) \geq \left(1 - \frac{2\varepsilon}{\rho}\right)^{m-1}. \quad (8)$$

In particular, if $\alpha \geq \delta\sqrt{\frac{m-1}{2m}}$, then $\rho = \delta/\sqrt{2}$ and $S(\varepsilon) \geq (1 - 2\sqrt{2}\varepsilon/\delta)^{m-1}$.

Proof. By (3), $S(\varepsilon) \geq \widehat{S}_2(\varepsilon)$. The case $\varepsilon = 0$ is immediate, so assume $\varepsilon > 0$.

Fix a cell $V_{\text{ELQ}}(\mathbf{q})$ and set $r = \rho$. By Proposition 3, $\mathbf{q}$ is at distance at least r from the boundary of the cell. Consider any convex polytope $P \subset \mathbb{R}^{m-1}$ containing a point c at distance at least r from ∂P . Scale P toward c by the factor $1 - 2\varepsilon/r$:

$$T(x) = c + \left(1 - \frac{2\varepsilon}{r}\right)(x - c).$$

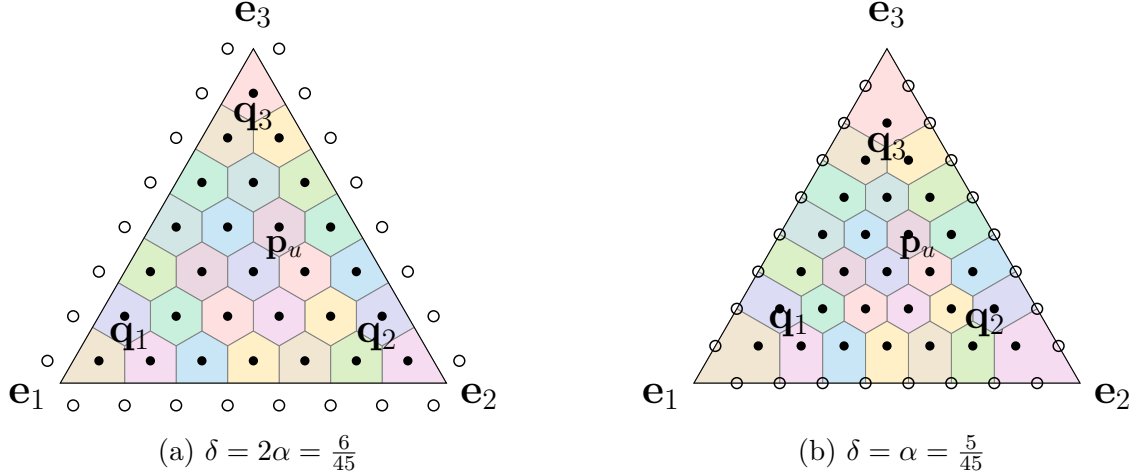


Figure 2: ELQ (ℓ_2 -Voronoi) cell partitions at $m = 3$, $k = 2$ ($K = 9$, 28 codewords). In (a) $\alpha = \delta/2$; lattice-adjacent points to the codebook lie outside the simplex. In (b), the same lattice-adjacent points lie on the simplex boundary.

We show that every point of $T(P)$ is at least 2ε from ∂P .

Since $B(c, r) \subseteq P$, set $\lambda = 2\varepsilon/r \in (0, 1]$. For any $z = c + (1 - \lambda)(x - c) \in T(P)$ and any w with $\|w\| \leq 2\varepsilon$, let $\tilde{w} = w/\lambda$. Then $\|\tilde{w}\| \leq r$, so $c + \tilde{w} \in P$, and

$$z + w = (1 - \lambda)x + \lambda(c + \tilde{w}).$$

This is a convex combination of two points in P , so $z + w \in P$. Hence every point of $T(P)$ is at least 2ε from ∂P .

Scaling by $1 - 2\varepsilon/r$ multiplies the $(m - 1)$ -dimensional volume by $(1 - 2\varepsilon/r)^{m-1}$. Therefore,

$$\text{Vol}(\{x \in P : \text{dist}(x, \partial P) \geq 2\varepsilon\}) \geq \left(1 - \frac{2\varepsilon}{r}\right)^{m-1} \text{Vol}(P).$$

Applying this to every ELQ cell with $c = \mathbf{q}$ gives $\widehat{S}_2(\varepsilon) \geq (1 - 2\varepsilon/\rho)^{m-1}$. $\square$

4.3 KL distortion bound

We next bound $D(\mathbf{p} \parallel \phi_{\text{ELQ}}(\mathbf{p}))$. We first consider lattice cells that are fully contained in the simplex, and then give a bound that applies to every $\mathbf{p} \in \Delta_{m-1}$.

Lemma 5 (KL bound under unclipping). *Let $\mathbf{q} \in \mathcal{C}_\delta$ and let $V_{\text{lat}}(\mathbf{q})$ denote its Voronoi cell with respect to the unrestricted lattice $\mathcal{L}_\delta \supset \mathcal{C}_\delta$. If $V_{\text{lat}}(\mathbf{q}) \subseteq \Delta_{m-1}$, then $\min_i q_i \geq \delta(m - 1)/m$, and for every $\mathbf{p} \in V_{\text{lat}}(\mathbf{q})$,*

$$D(\mathbf{p} \parallel \mathbf{q}) \leq \frac{m\delta^2}{4 \max(\alpha, \delta(m - 1)/m)}. \quad (9)$$

Proof. Recall from Equation (7) that the vertices of $V_{\text{lat}}(\mathbf{q})$ are $\mathbf{q} + \delta\sigma(\mathbf{u}^{(j)})$ with $\|\mathbf{u}^{(j)}\|_2^2 = j(m-j)/m \leq m/4$.

The unclipping assumption requires $\mathbf{q} + \delta\mathbf{v} \in \Delta_{m-1}$ for every Voronoi vertex $\delta\mathbf{v}$. The most negative single entry over all such $\mathbf{v}$ is $-(m-1)/m$, attained at $\mathbf{u}^{(1)}$; cycling this entry through each coordinate position via permutations forces $q_i \geq \delta(m-1)/m$ for all i , hence $\min_i q_i \geq \max(\alpha, \delta(m-1)/m)$ when combined with the codebook bound. The inequalities $D(\mathbf{p}||\mathbf{q}) \leq \sum_i (p_i - q_i)^2/q_i \leq \|\mathbf{p} - \mathbf{q}\|_2^2 / \min_i q_i$, together with the lattice covering bound $\|\mathbf{p} - \mathbf{q}\|_2^2 \leq m\delta^2/4$ stated above, yield the stated result. $\square$

Lemma 6 (Global ELQ bound). *For every $\mathbf{p} \in \Delta_{m-1}$ and $\mathbf{q} = \phi_{\text{ELQ}}(\mathbf{p})$ with any valid α and δ ,*

$$\|\mathbf{p} - \mathbf{q}\|_2 \leq \frac{1}{2}\delta\sqrt{m} + \alpha\sqrt{m(m-1)}. \quad (10)$$

Consequently, using $\min_i q_i \geq \alpha$,

$$D(\mathbf{p}||\mathbf{q}) \leq \frac{\left(\frac{1}{2}\delta\sqrt{m} + \alpha\sqrt{m(m-1)}\right)^2}{\alpha}. \quad (11)$$

Proof. We construct a specific $\tilde{\mathbf{q}} \in \mathcal{C}_\delta$ near $\mathbf{p}$; since ELQ chooses the nearest codeword in ℓ_2 distance, the same bound then holds for $\phi_{\text{ELQ}}(\mathbf{p})$.

Step 1 (move into the interior). Set $\mathbf{x} = (1 - m\alpha)\mathbf{p} + \alpha\mathbf{1}$. Then $\mathbf{x} \in \Delta_{m-1}$ with $\min_i x_i \geq \alpha$, and

$$\|\mathbf{x} - \mathbf{p}\|_2 = \alpha \|\mathbf{1} - m\mathbf{p}\|_2 \leq \alpha\sqrt{m(m-1)},$$

since $\|\mathbf{1} - m\mathbf{p}\|_2^2$ is convex in $\mathbf{p}$ and attains its maximum on Δ_{m-1} at a vertex $\mathbf{e}_j$, where $\|\mathbf{1} - m\mathbf{e}_j\|_2^2 = (m-1) + (m-1)^2 = m(m-1)$.

Step 2 (round to $\mathcal{C}_\delta$). By the $\mathbf{w}$ -parametrization of the codebook in (4), every $\tilde{\mathbf{q}} \in \mathcal{C}_\delta$ has the form $\tilde{\mathbf{q}} = \alpha\mathbf{1} + \delta\mathbf{w}$ with $\mathbf{w} \in \mathbb{Z}_{\geq 0}^m$ and $\sum_i w_i = km$. Define $\mathbf{y} \triangleq (\mathbf{x} - \alpha\mathbf{1})/\delta$; since $x_i \geq \alpha$ for every i , $\mathbf{y} \in \mathbb{R}_{\geq 0}^m$ and $\sum_i y_i = (1 - m\alpha)/\delta = km$. The ℓ_2 -nearest integer vector with $\sum_i w_i = km$ is obtained by the standard nearest-type rounding rule [10]: take the floors of the coordinates of $\mathbf{y}$ and assign the remaining counts to the largest residues. Since $y_i \geq 0$ for all i , the resulting vector also satisfies $w_i \geq 0$. Hence $\tilde{\mathbf{q}} \triangleq \alpha\mathbf{1} + \delta\mathbf{w}$ belongs to $\mathcal{C}_\delta$ and is a nearest point of the unrestricted lattice to $\mathbf{x}$, so $\mathbf{x} \in V_{\text{lat}}(\tilde{\mathbf{q}})$. The lattice covering bound above gives

$$\|\mathbf{x} - \tilde{\mathbf{q}}\|_2 \leq \frac{1}{2}\delta\sqrt{m}.$$

Step 3 (combine). By the triangle inequality and Steps 1-2,

$$\|\mathbf{p} - \tilde{\mathbf{q}}\|_2 \leq \|\mathbf{p} - \mathbf{x}\|_2 + \|\mathbf{x} - \tilde{\mathbf{q}}\|_2 \leq \alpha\sqrt{m(m-1)} + \frac{1}{2}\delta\sqrt{m}.$$

Since $\mathbf{q} = \phi_{\text{ELQ}}(\mathbf{p})$ is the ℓ_2 -nearest point of $\mathcal{C}_\delta$ to $\mathbf{p}$, the same bound holds with $\tilde{\mathbf{q}}$ replaced by $\mathbf{q}$, which proves the first claim. For the KL inequality, $D(\mathbf{p}||\mathbf{q}) \leq \sum_i (p_i - q_i)^2/q_i \leq \|\mathbf{p} - \mathbf{q}\|_2^2 / \min_i q_i$, and every codeword in $\mathcal{C}_\delta$ satisfies $\min_i q_i \geq \alpha$. Substituting the ℓ_2 bound gives the stated KL bound. $\square$

Reznik [10] derived an asymptotic $O(n^{-2})$ KL-distortion result away from the simplex boundary, with worse distortion near the boundary. Lemma 5 gives an explicit nonasymptotic bound with the same rate for unclipped cells bounded away from the boundary, while Lemma 6 gives a separate bound that applies over the full simplex.

4.4 α -selection heuristic

The symmetric choice $\alpha = \delta = 1/K$ simplifies the analysis, but need not minimize the distortion. Reznik suggested $\beta = 1/m$ for L_r -norm quantization, which corresponds to $\alpha/\delta = 1/m$ in our notation [10]. Here we choose α using a boundary–interior KL balance. Appendix A gives a local condition under which increasing α reduces the KL distortion away from the boundary. At the symmetric choice, numerical enumeration for $m \in \{2, 3, 4\}$ and $K \leq 20$ shows that the worst-case ELQ distortion is attained at a simplex vertex $\mathbf{p} = \mathbf{e}_j$. For general α , $\phi_{\text{ELQ}}(\mathbf{e}_j) = \mathbf{q}_j = (\alpha, \dots, \alpha, 1 - (m - 1)\alpha, \alpha, \dots, \alpha)$, giving distortion

$$B(\alpha) \triangleq D(\mathbf{e}_j \| \mathbf{q}_j) = \ln \frac{1}{1 - (m - 1)\alpha}. \quad (12)$$

$B(\alpha)$ is strictly increasing in α , so a smaller α gives lower distortion at the simplex vertices. To estimate the interior KL distortion, consider a Voronoi vertex of the corner cell, $\mathbf{p}^* = \mathbf{q}_j + \delta \mathbf{u}^{(1)}$, with the negative entry of $\mathbf{u}^{(1)} = (1/m, \dots, 1/m, -(m - 1)/m)$ aligned with the large coordinate of $\mathbf{q}_j$. A second-order expansion in δ gives

$$D(\mathbf{p}^* \| \mathbf{q}_j) \approx \frac{\delta^2}{2m^2} \left(\frac{m - 1}{\alpha} + \frac{(m - 1)^2}{1 - (m - 1)\alpha} \right).$$

In this second-order approximation, for small α , the first term dominates, so $D(\mathbf{p}^* \| \mathbf{q}_j) \approx \frac{(m-1)\delta^2}{2m^2\alpha}$. Setting this interior approximation equal to $B(\alpha) \approx (m - 1)\alpha$ gives $\delta \approx m\sqrt{2}\alpha$. Together with $\delta = (1/m - \alpha)/k$, this gives $\alpha \approx 1/[m(1 + \sqrt{2}mk)]$. For large mk , this simplifies to

$$\alpha^*(m, k) \approx \frac{1}{m^2 k \sqrt{2}}. \quad (13)$$

For $m = 3$, this gives $\alpha^* \approx 1/(9\sqrt{2}k) \approx 0.0786/k$, which is smaller than the symmetric choice $\alpha = \delta = 1/[m(k + 1)]$.

5 KL Quantizer (KLQ)

Since KL divergence is our measure of quantization distortion, we also consider a quantizer that directly minimizes it. For the symmetric codebook $\alpha = \delta = 1/K$ and $p_i > 0$ for all i , this becomes the K -type approximation problem studied by Geiger and Böcherer [11]. Their work focuses on finding the KL-optimal type efficiently, while here we use the lattice structure to study the KLQ cells and their smoothness.

5.1 Definition and convex cells

Definition 7. *The KL Quantizer is*

$$\phi_{\text{KL}}(\mathbf{p}) \in \arg \min_{\mathbf{q} \in \mathcal{C}_\delta} D(\mathbf{p} \| \mathbf{q}),$$

with ties broken arbitrarily. For the analysis, we use the closed KL Voronoi regions $V_{\text{KL}}(\mathbf{q}) = \{\mathbf{p} \in \Delta_{m-1} : D(\mathbf{p} \| \mathbf{q}) \leq D(\mathbf{p} \| \mathbf{q}') \ \forall \mathbf{q}' \in \mathcal{C}_\delta\}$. The tie-breaking choice only affects their shared boundaries.

Although $D(\mathbf{p}||\mathbf{q})$ is nonlinear in $\mathbf{p}$, the difference $D(\mathbf{p}||\mathbf{q}) - D(\mathbf{p}||\mathbf{q}')$ is linear in $\mathbf{p}$, so the cell boundaries are hyperplanes.

Lemma 8. *For $\mathbf{q}, \mathbf{q}' \in \mathcal{C}_\delta$, the set $\{\mathbf{p} \in \Delta_{m-1} : D(\mathbf{p}||\mathbf{q}) \leq D(\mathbf{p}||\mathbf{q}')\}$ is the intersection of Δ_{m-1} with a half-space. Thus, every $V_{\text{KL}}(\mathbf{q})$ is a convex polytope.*

Proof. $D(\mathbf{p}||\mathbf{q}) - D(\mathbf{p}||\mathbf{q}') = \sum_i p_i \ln(q'_i/q_i)$ after the $p_i \ln p_i$ terms cancel. This is linear in $\mathbf{p}$. $\square$

KL cells from local comparisons. The KL objective over the integer coordinates is a separable convex allocation problem of the type studied by Geiger and Böcherer [11]. We use this structure to show that it is enough to compare with lattice-adjacent codewords when defining a KLQ cell.

By Lemma 1, every codeword $\mathbf{q} \in \mathcal{C}_\delta$ has a unique representation $\mathbf{q} = \alpha \mathbf{1} + \delta \mathbf{w}$ where $\mathbf{w} \in \mathbb{Z}_{\geq 0}^m$ with $\sum_i w_i = km$; we call $\mathbf{w}$ the *integer coordinate* of $\mathbf{q}$, and identify $\mathbf{q}$ with $\mathbf{w}$ when convenient. A *lattice-adjacent* swap from $\mathbf{q}$ is the codeword $\mathbf{q}' = \mathbf{q} + \delta(\mathbf{e}_a - \mathbf{e}_b)$ whose integer coordinate is $\mathbf{w}' = \mathbf{w} + \mathbf{e}_a - \mathbf{e}_b$; this is well-defined whenever $w_b \geq 1$ (so that $\mathbf{w}' \geq \mathbf{0}$). Write $\mathcal{N}(\mathbf{q}) \subset \mathcal{C}_\delta$ for the set of such neighbours.

Lemma 9 (Adjacent comparisons determine the KL cell). *For every $\mathbf{q} \in \mathcal{C}_\delta$,*

$$V_{\text{KL}}(\mathbf{q}) = \{\mathbf{p} \in \Delta_{m-1} : D(\mathbf{p}||\mathbf{q}) \leq D(\mathbf{p}||\mathbf{q}') \ \forall \mathbf{q}' \in \mathcal{N}(\mathbf{q})\}.$$

Proof. The inclusion $V_{\text{KL}}(\mathbf{q}) \subseteq \{\dots\}$ is immediate. For the reverse, write $\mathbf{q} = \alpha \mathbf{1} + \delta \mathbf{w}$ and define $g_i(t) = -p_i \ln(\alpha + \delta t)$ and $f(\mathbf{v}) = \sum_i g_i(v_i)$. Minimizing $D(\mathbf{p}||\mathbf{q})$ over $\mathcal{C}_\delta$ is equivalent to minimizing $f(\mathbf{w})$ over $\mathbf{w} \in \mathbb{Z}_{\geq 0}^m$ with $\sum_i w_i = km$. Each g_i is convex, since $g_i''(t) = p_i \delta^2 / (\alpha + \delta t)^2 \geq 0$.

Suppose no lattice-adjacent swap decreases f . Then, for every i, j with $w_j \geq 1$,

$$g_i(w_i + 1) - g_i(w_i) \geq g_j(w_j) - g_j(w_j - 1).$$

By the separable-convex allocation property [11], this condition implies that $\mathbf{w}$ is a global minimizer of f under the fixed-sum constraint. Hence $D(\mathbf{p}||\mathbf{q}) \leq D(\mathbf{p}||\mathbf{q}')$ for every $\mathbf{q}' \in \mathcal{C}_\delta$, so $\mathbf{p} \in V_{\text{KL}}(\mathbf{q})$. $\square$

Remark 10 (KLQ computation). *Geiger and Böcherer give an exact greedy algorithm that allocates counts one at a time for this separable convex problem [11]. Lemma 9 gives an alternative local-search view, where counts are moved between coordinates by lattice-adjacent swaps until no improving swap remains. The resulting codeword is KL-optimal.*

Figure 3 shows the two KLQ cell partitions at $m = 3$, $k = 2$ ($K = 9$). The codebook is the same as ELQ at the corresponding α , but the cells differ because KL-bisectors differ from ℓ_2 -bisectors.

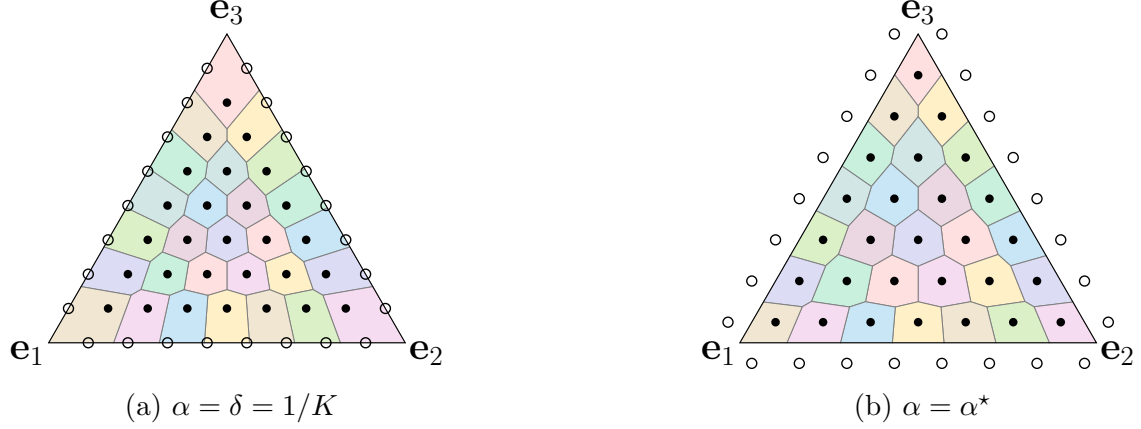


Figure 3: KLQ (KL-Voronoi) cell partitions at $m = 3$, $k = 2$ ($K = 9$, 28 codewords). Cells are convex polytopes (Lemma 8).

5.2 Inradius and smoothness

By Lemma 9, only lattice-adjacent KL bisectors need to be considered when finding the inradius.

Definition 11. For $\mathbf{q} \in \mathcal{C}_\delta$, define $\rho(\mathbf{q}) = \text{dist}(\mathbf{q}, \partial V_{\text{KL}}(\mathbf{q}))$, where the distance and boundary are taken in the affine hull of Δ_{m-1} .

The nearest boundary of a KLQ cell can come from either a lattice-adjacent KL bisector or a simplex facet.

Proposition 12 (KLQ inradius at $\alpha = \delta = 1/K$). At $\alpha = \delta = 1/K$, with $K = m(k+1)$, the KLQ inradius satisfies

$$\rho(m, K) \triangleq \min_{\mathbf{q} \in \mathcal{C}_\delta} \rho(\mathbf{q}) \geq \frac{1 - \ln 2}{\ln 2} \sqrt{\frac{m}{m-1}} \cdot \frac{1}{K}, \quad (14)$$

and the bound is asymptotically tight as $k \rightarrow \infty$ (equivalently, $K \rightarrow \infty$ along $K = m(k+1)$).

Proof sketch. At $\alpha = \delta = 1/K$, write $u_i \triangleq Kq_i = w_i + 1 \geq 1$. By Lemmas 8 and 9, $\rho(\mathbf{q})$ is the smaller of the distance to a lattice-adjacent KL bisector and the distance to a simplex facet. The calculation in Appendix B shows that every adjacent-bisector distance is at least

$$\frac{1 - \ln 2}{\ln 2} \sqrt{\frac{m}{m-1}} \frac{1}{K},$$

and corner codewords approach this bound as $k \rightarrow \infty$. The simplex-facet distance is at least $(1/K)\sqrt{m/(m-1)}$, which is larger because $(1 - \ln 2)/\ln 2 < 1$. $\square$

Theorem 13 (KLQ smoothness). At $\alpha = \delta = 1/K$, with $K = m(k+1)$, for all $\varepsilon \in [0, \rho/2]$,

$$S(\varepsilon) \geq \widehat{S}_2(\varepsilon) \geq (1 - 2\varepsilon/\rho)^{m-1}, \quad \rho = \min_{\mathbf{q} \in \mathcal{C}_\delta} \rho(\mathbf{q}).$$

Proof. The proof uses the same scaling argument as Theorem 4. KLQ cells are convex by Lemma 8, and Proposition 12 gives the required inradius bound. $\square$

5.3 Worst-case KL distortion

In this subsection, we take $\alpha = \delta = 1/K$ with $K = m(k+1)$, so that $\mathcal{C}_\delta = \mathring{\mathbb{S}}_K^m$. For each $\mathbf{p} \in \Delta_{m-1}$, we construct a codeword $\mathbf{q}_0 \in \mathcal{C}_\delta$ with a bound on $\max_{i:p_i>0} p_i/q_{0,i}$. KL-optimality then gives the corresponding bound for $\phi_{\text{KL}}(\mathbf{p})$.

Theorem 14. *For every boundary type $\hat{\mathbf{p}} \in \mathring{\mathbb{S}}_K^m$,*

$$R^*(\hat{\mathbf{p}}) \triangleq \min_{\mathbf{q} \in \mathring{\mathbb{S}}_K^m} \max_{i:\hat{p}_i>0} \frac{\hat{p}_i}{q_i} \leq \frac{K}{K-m+1}, \quad (15)$$

with equality at $\hat{\mathbf{p}} = \mathbf{e}_1$, attained by $\mathbf{q}^ = \frac{1}{K}(K-m+1, 1, \dots, 1)$.*

Proof. Set $R = K/(K-m+1)$ and write any type $\mathbf{q} \in \mathring{\mathbb{S}}_K^m$ via its count vector $\mathbf{u} \triangleq K\mathbf{q} \in \mathbb{Z}_{\geq 1}^m$ with $\sum_i u_i = K$; similarly $\hat{\mathbf{v}} \triangleq K\hat{\mathbf{p}}$. Define

$$\tilde{u}_i = \begin{cases} 1 & \hat{v}_i = 0, \\ \lceil \hat{v}_i(K-m+1)/K \rceil & \hat{v}_i \geq 1. \end{cases}$$

Then $\tilde{u}_i \geq 1$. For $\hat{v}_i \geq 1$, $\tilde{u}_i \geq \hat{v}_i(K-m+1)/K$, so $\hat{v}_i/\tilde{u}_i \leq R$. Let $s = |\{i : \hat{v}_i \geq 1\}|$. Then

$$\begin{aligned} \sum_i \tilde{u}_i &< (m-s) + \sum_{\hat{v}_i \geq 1} (\hat{v}_i(K-m+1)/K + 1) \\ &= (m-s) + (K-m+1) + s = K+1. \end{aligned}$$

Since $\sum_i \tilde{u}_i$ is an integer, $\sum_i \tilde{u}_i \leq K$. Distribute the remaining $K - \sum_i \tilde{u}_i$ counts among the coordinates; this only decreases ratios. Set $\tilde{\mathbf{q}} = \tilde{\mathbf{u}}/K \in \mathring{\mathbb{S}}_K^m$. For $\hat{\mathbf{p}} = \mathbf{e}_1$, every $\mathbf{q} \in \mathring{\mathbb{S}}_K^m$ has $Kq_1 \leq K - (m-1)$, forcing $\hat{p}_1/q_1 \geq R$. $\square$

The same construction extends from boundary types to all $\mathbf{p} \in \Delta_{m-1}$.

Corollary 15. *For every $\mathbf{p} \in \Delta_{m-1}$, there exists $\mathbf{q} \in \mathcal{C}_\delta$ with $\max_{i:p_i>0} \frac{p_i}{q_i} \leq \frac{K}{K-m+1}$.*

Proof. Set $u_i = \max(1, \lceil p_i(K-m+1) \rceil)$. As in the proof of Theorem 14, $\sum_i u_i \leq K$ and $p_i K/u_i \leq K/(K-m+1)$ for every i with $p_i > 0$. Distribute the remaining $K - \sum_i u_i$ counts among the coordinates and set $\mathbf{q} = \mathbf{u}/K$. Increasing the counts can only decrease the ratios, so the same bound holds. $\square$

Using Corollary 15, Equation (1), and KL-optimality gives the following worst-case bound.

Theorem 16. *At $\alpha = \delta = 1/K$, with $K = m(k+1)$,*

$$\max_{\mathbf{p} \in \Delta_{m-1}} D(\mathbf{p} \parallel \phi_{\text{KL}}(\mathbf{p})) = \ln \frac{K}{K-m+1}.$$

Proof. By Corollary 15 there exists $\mathbf{q}_0 \in \mathcal{C}_\delta$ with $\max_{i:p_i>0} \frac{p_i}{q_{0,i}} \leq \frac{K}{K-m+1}$. By (1), $D(\mathbf{p} \parallel \mathbf{q}_0) \leq \ln(\max_{i:p_i>0} \frac{p_i}{q_{0,i}}) \leq \ln \frac{K}{K-m+1}$. By the KL-optimality of ϕ_{KL} ,

$$D(\mathbf{p} \parallel \phi_{\text{KL}}(\mathbf{p})) \leq D(\mathbf{p} \parallel \mathbf{q}_0) \leq \ln \frac{K}{K-m+1}.$$

Equality is attained at $\mathbf{p} = \mathbf{e}_1$, since every $\mathbf{q} \in \mathcal{C}_\delta$ satisfies $q_1 \leq (K-m+1)/K$, and this value is attained by $\mathbf{q}^* = (K-m+1, 1, \dots, 1)/K$. $\square$

For distributions with $p_i > 0$ for all i , this is the same K -type KL approximation problem studied by Geiger and Böcherer [11]. Their distribution-dependent bound can be tighter away from the simplex boundary. For example, when $p_i \geq 1/K$ for all i , they obtain $D(\mathbf{p} \parallel \phi_{\text{KL}}(\mathbf{p})) \leq m \ln 2 / (2K)$. As $\mathbf{p}$ remains in the simplex interior and approaches a vertex, their bound approaches $\ln \frac{K}{K-m+1} + \frac{\ln 2}{2(K-m+1)}$, while our distortion approaches $\ln \frac{K}{K-m+1}$, which is the exact worst-case distortion over the full simplex. Their proof uses a reverse I-projection and a variational-distance bound, while our exact result follows from a direct construction of a positive K -type and a matching lower bound at a simplex vertex.

Adler, Tang, and Polyanskiy studied a different quantizer based on entrywise companding and proved a worst-case KL bound of order $N_A^{-2} \log^2 m$, where N_A is the number of quantization levels per coordinate [12]. Their representation uses $m \lceil \log_2 N_A \rceil$ bits, while an index of the symmetric lattice codebook here requires $\lceil \log_2 \binom{K-1}{m-1} \rceil$ bits, so N_A and K should not be compared directly. Our use of the lattice is motivated by the smoothness analysis in Section 5.2, rather than KL distortion alone.

For $\mathbf{p}$ lying in a lattice Voronoi cell unclipped by the simplex boundary, KL-optimality of ϕ_{KL} together with Lemma 5 gives the tighter bound $D(\mathbf{p} \parallel \phi_{\text{KL}}(\mathbf{p})) \leq 1/[4(k+1)]$ (at $\alpha = \delta = 1/K$).

5.4 Lloyd algorithm and centroidal optimality

We also apply Lloyd's algorithm [14] to reduce the average KL distortion by alternating between a cell update and a centroid update.

Let $\mathbf{p} \sim \text{Unif}(\Delta_{m-1})$ and write $J(\mathcal{C}) = \mathbb{E}[\min_{\mathbf{q} \in \mathcal{C}} D(\mathbf{p} \parallel \mathbf{q})]$.

Lloyd iteration Given a codebook $\mathcal{C}^{(t)} = \{\mathbf{q}_1^{(t)}, \dots, \mathbf{q}_N^{(t)}\}$:

- *Cell update:* form $V_i^{(t)} = \{\mathbf{p} \in \Delta_{m-1} : D(\mathbf{p} \parallel \mathbf{q}_i^{(t)}) \leq D(\mathbf{p} \parallel \mathbf{q}_j^{(t)}) \ \forall j\}$.
- *Centroid update:* set $\mathbf{q}_i^{(t+1)} = \arg \min_{\mathbf{q} \in \Delta_{m-1}^\circ} \mathbb{E}[D(\mathbf{p} \parallel \mathbf{q}) \mid \mathbf{p} \in V_i^{(t)}]$.

For KL divergence, the minimizing codeword is the conditional mean of $\mathbf{p}$ in the cell, a standard result for Bregman divergences [15], so $\mathbf{q}_i^{(t+1)} = \mathbb{E}[\mathbf{p} \mid \mathbf{p} \in V_i^{(t)}]$.

Under the uniform distribution on Δ_{m-1} , this conditional mean is the geometric centroid of the cell.

Numerical observation Each Lloyd sweep is non-increasing in $J(\mathcal{C})$, so the sequence of objective values converges. Starting from $\mathcal{C}^{(0)} = \mathring{\mathbb{S}}_K^m$ at $m = 3$, $k = 3$ ($K = 12$), the iteration stops after about 70 sweeps. The smallest coordinate of the final codebook is 0.0275, close to the heuristic value $\alpha^* \approx 0.0262$ from Section 4.4. The Lloyd-optimized codebook is not restricted to the lattice family $\mathcal{C}_\delta$, so we use this only as a numerical comparison; the analytical KLQ bounds above apply to the lattice codebook.

6 Simulation

We evaluate the KL distortion of the studied quantizers and their effect on Huffman code stability. Source distributions are drawn from $\text{Dir}(1, \dots, 1)$ on alphabet size $m = 20$. For each source distribution, we generate an i.i.d. sequence of length $L = 1000$ and compute its empirical distribution $\mathbf{p}$. We then delete n_{del} symbols chosen uniformly at random and let $\mathbf{p}'$ denote the empirical distribution after deletion, with $n_{\text{del}} \in \{1, 2, 5, 10, 20, 50, 100\}$. We use 500 original sequences and repeat the random deletion three times for each value of n_{del} . We fix $k = 1$, so that $K = 2m = 40$, and compare ELQ and KLQ at both the symmetric lattice $\alpha = \delta$ and the heuristic choice $\alpha = \alpha^*$ of Section 4.4. We also compare with the approximate minimax compander (AMC) of Adler, Tang, and Polyanskiy [12]. We use $N_A = 4$ quantization levels per coordinate. At $m = 20$, this uses $20 \log_2 4 = 40$ bits, close to the 37 bits needed to index the $K = 40$ lattice codebook. AMC uses midpoint decoding followed by normalization [12]. ELQ, KLQ, and AMC first quantize $\mathbf{p}$ and then build a canonical Huffman code from the quantized distribution. We also include standard Huffman and canonical Huffman applied directly to $\mathbf{p}$.

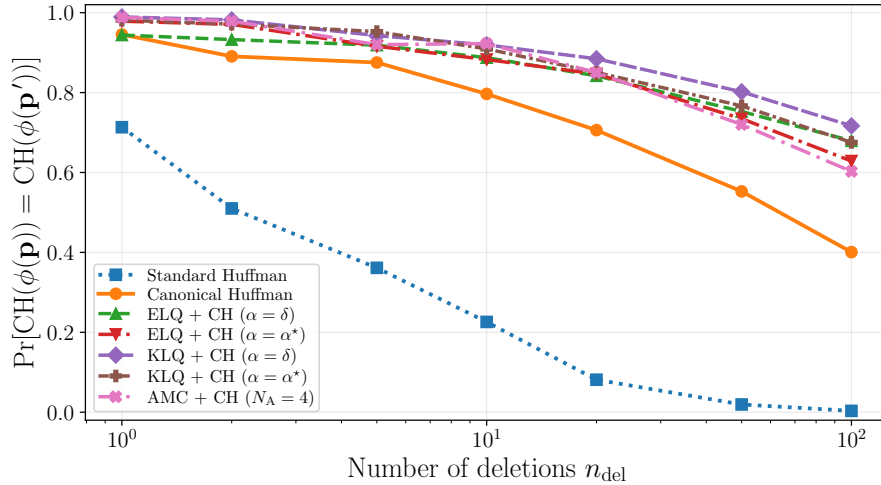


Figure 4: Code stability under random deletions at $k = 1$ ($K = 40$, $m = 20$, $L = 1000$). Standard and canonical Huffman are applied directly to the empirical distribution. ELQ, KLQ, and AMC first quantize the empirical distribution and then build a canonical Huffman code.

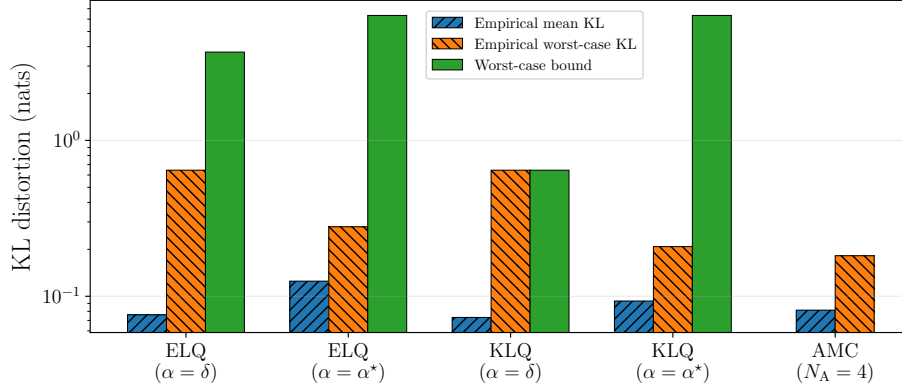


Figure 5: KL distortion at $k = 1$ ($K = 40$, $m = 20$, $L = 1000$). We report the empirical mean and an empirical worst case over sampled distributions and simplex vertices. Theoretical worst-case bounds are also shown for ELQ and KLQ.

For each trial, we record two quantities. First, we measure code stability under deletion: $\Pr[\text{CH}(\phi(\mathbf{p})) = \text{CH}(\phi(\mathbf{p}'))]$, where $\text{CH}(\cdot)$ denotes the canonical Huffman code built from the input distribution. For standard Huffman, we record whether the codebook remains unchanged after deletion. Second, we measure the KL distortion of the quantizer, $D(\mathbf{p} \parallel \phi(\mathbf{p}))$. Since this KL distortion depends only on the original empirical distribution and the quantizer, it is independent of n_{del} . For the empirical worst case in Figure 5, we test all simplex vertices and 5000 additional samples from $\text{Dir}(1, \dots, 1)$. For ELQ, we use the tighter of the global bound in Lemma 6 and the general bound $D(\mathbf{p} \parallel \mathbf{q}) \leq \ln(1/\alpha)$. For KLQ at $\alpha = \alpha^*$, the same bound applies by KL-optimality. For KLQ at $\alpha = \delta$, we use the exact worst-case bound in Theorem 16. For AMC, we report the empirical mean and empirical worst-case KL distortion. We do not plot its theoretical worst-case bound because the nonasymptotic bound in [12] does not apply at $m = 20$ and $N_A = 4$.

Figure 4 shows that quantization can make the canonical Huffman code more stable under random deletions. Figure 5 compares the corresponding KL distortion, including AMC as a related-work baseline. The empirical distortions are below the worst-case bounds in several settings, while the bounds still provide uniform guarantees.

7 Conclusion

We studied two lattice quantizers on the probability simplex: the Euclidean Lattice Quantizer (ELQ) and the KL Quantizer (KLQ). For both quantizers, we derived KL distortion bounds and smoothness guarantees under perturbations. For KLQ, we also showed that its KL Voronoi cells are determined by lattice-adjacent codewords, which leads to a local-search implementation. We further gave a simple heuristic choice $\alpha^* \approx 1/(m^2 k \sqrt{2})$ for the lattice offset. Simulations illustrate the trade-off between KL distortion and Huffman code stability, with AMC as a related-work baseline.

A A monotonicity condition for α -selection

This appendix gives a local monotonicity condition used in the α -selection discussion of Section 4.4.

Fix k and let $\mathcal{W} = \{\mathbf{w} \in \mathbb{Z}_{\geq 0}^m : \sum_i w_i = km\}$. For $0 < \alpha \leq 1/[m(k+1)]$, set $\delta = (1/m - \alpha)/k$ and $\mathcal{C}_\alpha = \{\alpha \mathbf{1} + \delta \mathbf{w} : \mathbf{w} \in \mathcal{W}\}$. For fixed $\mathbf{w}$, define $F_{\mathbf{w}}(\alpha; \mathbf{p}) = \sum_i p_i \ln q_i(\alpha; \mathbf{w})$, where $q_i(\alpha; \mathbf{w}) = \alpha + \delta w_i$, and let $F^*(\alpha; \mathbf{p}) = \max_{\mathbf{w} \in \mathcal{W}} F_{\mathbf{w}}(\alpha; \mathbf{p})$. Since $\min_{\mathbf{q} \in \mathcal{C}_\alpha} D(\mathbf{p} \parallel \mathbf{q}) = -H(\mathbf{p}) - F^*(\alpha; \mathbf{p})$, a larger F^* gives a smaller minimum KL distortion.

Lemma 17. *For fixed $\mathbf{w}$,*

$$\frac{d}{d\alpha} F_{\mathbf{w}}(\alpha; \mathbf{p}) = \frac{1}{1/m - \alpha} \left(\frac{1}{m} \sum_i \frac{p_i}{q_i} - 1 \right).$$

Proof. Since $\frac{dq_i}{d\alpha} = 1 - w_i/k$,

$$\frac{d}{d\alpha} F_{\mathbf{w}}(\alpha; \mathbf{p}) = \sum_i \frac{p_i}{q_i} \left(1 - \frac{w_i}{k} \right).$$

Using $1 - w_i/k = (1/m - q_i)/(1/m - \alpha)$ and $\sum_i p_i = 1$ gives the result. $\square$

Proposition 18. *Fix $\mathbf{p} \in \Delta_{m-1}$ and $0 < \alpha_0 < 1/[m(k+1)]$. Let $\mathbf{w}^*(\alpha_0) \in \arg \max_{\mathbf{w} \in \mathcal{W}} F_{\mathbf{w}}(\alpha_0; \mathbf{p})$. If*

$$\frac{1}{m} \sum_i \frac{p_i}{q_i(\alpha_0; \mathbf{w}^*(\alpha_0))} > 1,$$

then, for all sufficiently small $h > 0$, $F^(\alpha_0 + h; \mathbf{p}) > F^*(\alpha_0; \mathbf{p})$. Hence the minimum KL distortion over $\mathcal{C}_\alpha$ decreases locally as α increases from α_0 .*

Proof. Let $\mathbf{w}_0 = \mathbf{w}^*(\alpha_0)$. The lemma gives $\frac{d}{d\alpha} F_{\mathbf{w}_0}(\alpha_0; \mathbf{p}) > 0$, so for sufficiently small $h > 0$, $F_{\mathbf{w}_0}(\alpha_0 + h; \mathbf{p}) > F_{\mathbf{w}_0}(\alpha_0; \mathbf{p})$. Therefore

$$F^*(\alpha_0 + h; \mathbf{p}) \geq F_{\mathbf{w}_0}(\alpha_0 + h; \mathbf{p}) > F_{\mathbf{w}_0}(\alpha_0; \mathbf{p}) = F^*(\alpha_0; \mathbf{p}).$$

$\square$

By Cauchy-Schwarz, $\sum_i p_i/q_i \geq 1/\langle \mathbf{p}, \mathbf{q} \rangle$. Hence $\langle \mathbf{p}, \mathbf{q} \rangle < 1/m$ is a sufficient condition for the derivative in the lemma to be positive. This condition does not hold for every nonuniform $\mathbf{p}$ and $\mathbf{q}$, so the proposition gives a local condition rather than global monotonicity in α .

On the other hand, the vertex distortion $B(\alpha) = \ln(1/(1 - (m-1)\alpha))$ is strictly increasing in α . Thus increasing α can reduce the distortion away from the boundary under the condition above, while increasing the distortion at simplex vertices. This is the trade-off used in Section 4.4.

B KLQ inradius

We give the details for Proposition 12.

Lemma 19 (Inradius formula). *Let $\mathbf{n}_H(\mathbf{q}, \mathbf{q}')$ denote the projection onto $H_0 = \{\mathbf{v} \in \mathbb{R}^m : \sum_i v_i = 0\}$ of the vector $(\ln(q'_i/q_i))_i$. Then*

$$\rho(\mathbf{q}) = \min \left\{ \min_{\mathbf{q}' \neq \mathbf{q}} \frac{D(\mathbf{q} \parallel \mathbf{q}')}{\|\mathbf{n}_H(\mathbf{q}, \mathbf{q}')\|_2}, \min_i q_i \sqrt{\frac{m}{m-1}} \right\}.$$

Proof. We work in the affine hyperplane $H = \{\mathbf{x} \in \mathbb{R}^m : \sum_i x_i = 1\}$ and write $\mathbf{p} = \mathbf{q} + \mathbf{v}$ with $\mathbf{v} \in H_0$.

For $\mathbf{q}' \neq \mathbf{q}$, let $a_i = \ln(q'_i/q_i)$. By Lemma 8, the KL bisector between $\mathbf{q}$ and $\mathbf{q}'$ satisfies $\sum_i p_i a_i = 0$. Substituting $\mathbf{p} = \mathbf{q} + \mathbf{v}$ gives $\sum_i v_i a_i = D(\mathbf{q} \parallel \mathbf{q}')$. Since $\mathbf{v} \in H_0$, the left-hand side is $\langle \mathbf{n}_H(\mathbf{q}, \mathbf{q}'), \mathbf{v} \rangle$. Hence the distance from $\mathbf{q}$ to this bisector is $D(\mathbf{q} \parallel \mathbf{q}') / \|\mathbf{n}_H(\mathbf{q}, \mathbf{q}')\|_2$.

For the simplex facet $\{p_i = 0\} \cap H$, the normal within H is $\mathbf{e}_i - \mathbf{1}/m$, whose norm is $\sqrt{(m-1)/m}$. Thus the distance from $\mathbf{q}$ to this facet is $q_i \sqrt{m/(m-1)}$. Taking the minimum over the bisectors and simplex facets gives the result. $\square$

Corollary 20 (Adjacent-bisector reduction). *The inradius can be written as*

$$\rho(\mathbf{q}) = \min \left\{ \min_{\mathbf{q}' \in \mathcal{N}(\mathbf{q})} \frac{D(\mathbf{q} \parallel \mathbf{q}')}{\|\mathbf{n}_H(\mathbf{q}, \mathbf{q}')\|_2}, \min_i q_i \sqrt{\frac{m}{m-1}} \right\}.$$

Proof. By Lemma 9, the KL cell is determined by the lattice-adjacent bisector half-spaces together with the simplex facets. The inradius is therefore the minimum distance from $\mathbf{q}$ to these hyperplanes. $\square$

Lemma 21 (Inradius lower bound at $\alpha = \delta = 1/K$). *Let $K = m(k+1)$ with $m \geq 2$ and $k \geq 1$. At $\alpha = \delta = 1/K$,*

$$\rho(m, K) \triangleq \min_{\mathbf{q} \in \mathcal{C}_\delta} \rho(\mathbf{q}) \geq \frac{1 - \ln 2}{\ln 2} \sqrt{\frac{m}{m-1}} \frac{1}{K}.$$

For fixed m , the bound is asymptotically tight as $k \rightarrow \infty$. In particular, $\rho(m, K) = \Theta(1/K)$.

Proof. At $\alpha = \delta = 1/K$, write $q_i = r_i/K$, where $r_i \geq 1$ and $\sum_i r_i = K$. By Corollary 20, it is enough to consider lattice-adjacent codewords $\mathbf{q}' = \mathbf{q} + (\mathbf{e}_a - \mathbf{e}_b)/K$. For this move, $r_b \geq 2$.

Let $A = \ln(1 + 1/r_a)$ and $B = \ln(1 + 1/(r_b - 1))$. Then $D(\mathbf{q} \parallel \mathbf{q}') = (r_b B - r_a A)/K$. The log-ratio vector has entries A and $-B$ in coordinates a and b , so $\|\mathbf{n}_H(\mathbf{q}, \mathbf{q}')\|_2^2 = A^2 + B^2 - (A - B)^2/m$. Therefore the distance to this bisector is

$$d(r_a, r_b) = \frac{1}{K} \frac{r_b B - r_a A}{\sqrt{A^2 + B^2 - (A - B)^2/m}}.$$

For fixed r_a , letting $r_b \rightarrow \infty$ gives $B \rightarrow 0$ and $r_b B \rightarrow 1$. Hence

$$\lim_{r_b \rightarrow \infty} d(r_a, r_b) = \frac{1}{K} h(r_a) \sqrt{\frac{m}{m-1}}, \quad h(r) = \frac{1 - r \ln(1 + 1/r)}{\ln(1 + 1/r)}.$$

The function h is increasing. Indeed, if $A_r = \ln(1 + 1/r)$, then $h(r) = 1/A_r - r$ and $h'(r) = 1/[r(r+1)A_r^2] - 1$. Also, $\ln(1 + 1/r) < 1/\sqrt{r(r+1)}$ for $r > 0$. To see this, set $z = \sqrt{1 + 1/r} > 1$; the inequality becomes $2 \ln z < z - z^{-1}$, and $z - z^{-1} - 2 \ln z$ is zero at $z = 1$ and strictly increasing for $z > 1$. Thus $h'(r) > 0$. Since $h(1) = (1 - \ln 2)/\ln 2$ and $h(r) \rightarrow 1/2$, we have $0 < h(r) < 1/2$ for finite $r \geq 1$.

It remains to compare finite r_b with the limit above. Let $\beta = h(r_a)$ and set $s = B$. Since $e^s = r_b/(r_b - 1)$, we have $r_b B = s/(1 - e^{-s})$. For $s > 0$, $s/(1 - e^{-s}) \geq 1 + s/2$. This is equivalent to $(s - 2)e^s + s + 2 \geq 0$; if $L(s) = (s - 2)e^s + s + 2$, then $L(0) = L'(0) = 0$ and $L''(s) = se^s \geq 0$.

Since $1 - r_a A = A\beta$, it follows that $r_b B - r_a A \geq A\beta + s/2$. It is therefore enough to show

$$\frac{A\beta + s/2}{\sqrt{A^2 + s^2 - (A - s)^2/m}} \geq \beta \sqrt{\frac{m}{m-1}}.$$

Squaring and simplifying reduces this to

$$s \left[A\beta \left(1 - \frac{2\beta}{m-1} \right) + s \left(\frac{1}{4} - \beta^2 \right) \right] \geq 0.$$

Since $0 < \beta < 1/2$ and $m \geq 2$, both terms in brackets are nonnegative. Thus

$$d(r_a, r_b) \geq \frac{1}{K} h(r_a) \sqrt{\frac{m}{m-1}} \geq \frac{1}{K} \frac{1 - \ln 2}{\ln 2} \sqrt{\frac{m}{m-1}}.$$

The distance to any simplex facet is at least $(1/K)\sqrt{m/(m-1)}$. Since $(1 - \ln 2)/\ln 2 < 1$, the facet term also satisfies the claimed lower bound.

Finally, consider a codeword with $r_a = 1$, $r_b = K - m + 1$, and all other counts equal to 1. As $k \rightarrow \infty$, the distance to the adjacent bisector obtained by moving one count from b to a approaches

$$\frac{1}{K} \frac{1 - \ln 2}{\ln 2} \sqrt{\frac{m}{m-1}}.$$

Since the inradius cannot exceed this bisector distance, the lower bound is asymptotically tight. $\square$

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